

A

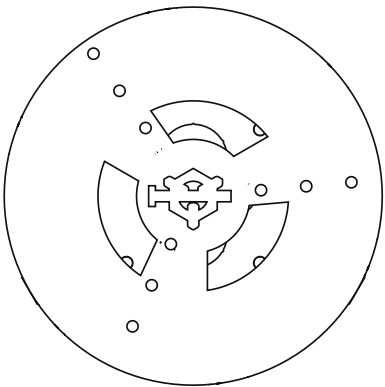
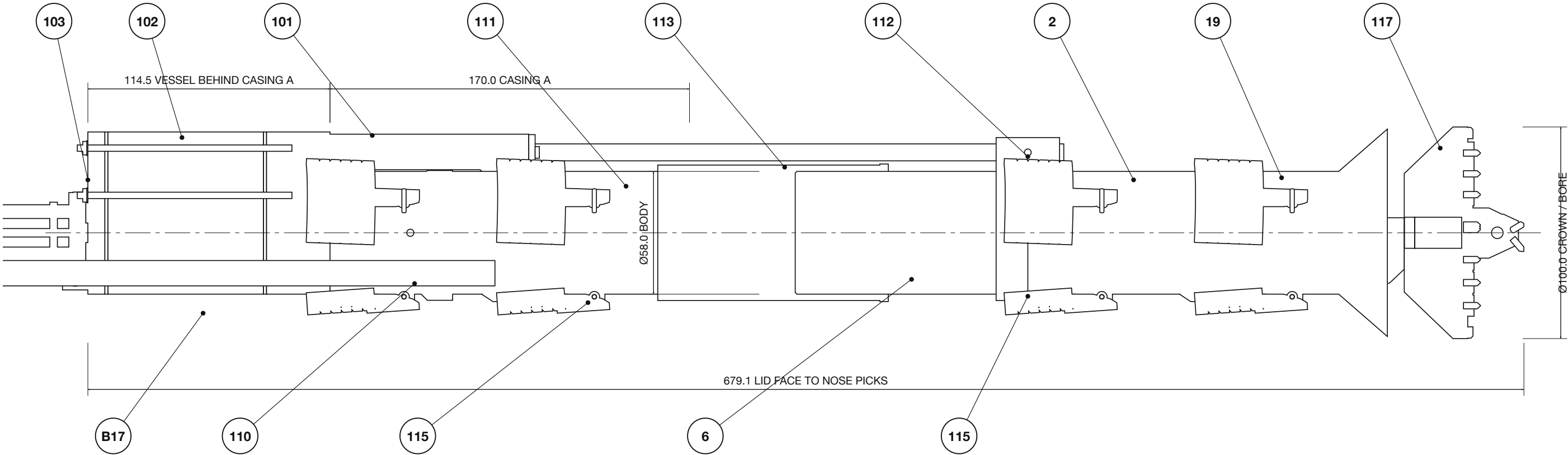
B

C

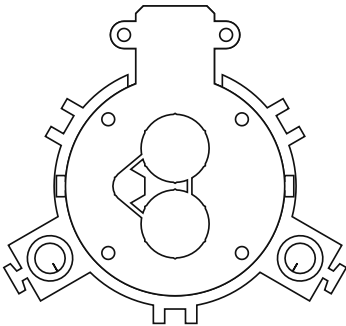
D

GENERAL ARRANGEMENT · SIDE ELEVATION

SCALE 1:2 · SHOES AT REST (4°) · B SECTION FULLY EXTENDED



VIEW FROM FRONT · 1:2



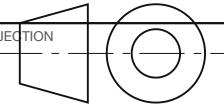
VIEW FROM REAR · 1:2

LID, 2 × WETLINK M10, M5 AIR FITTING

KEY DATA

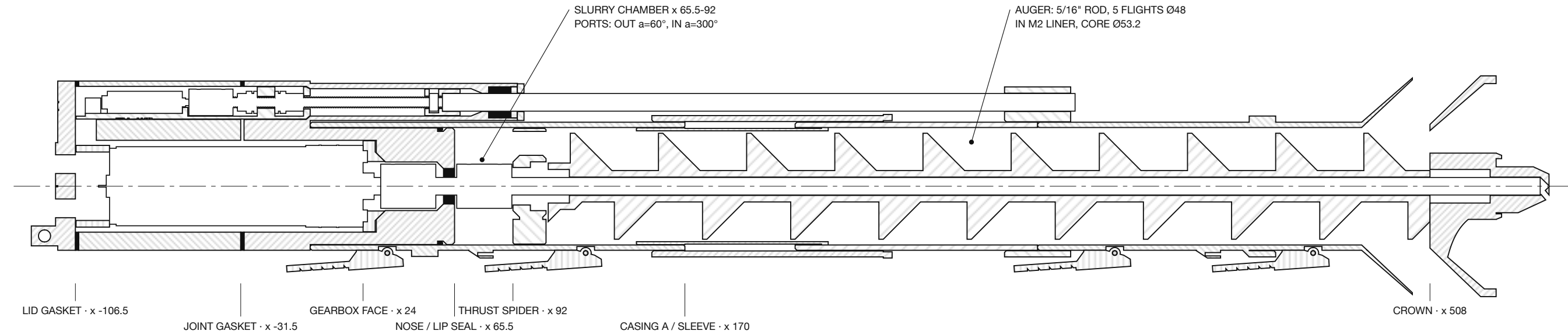
Bore (crown gauge)	Ø100
Stroke	50 (hard stop 55)
Worm length, lid to nose	679.1
Estimated mass	1.6-1.9 kg
Printed parts	26 designs
Seals on the dry vessel	7
Largest print	124 mm tall

UNLESS OTHERWISE STATED: DIMENSIONS IN mm · PRINTED PARTS ±0.2 · DO NOT SCALE THE DRAWING
PRINTED PARTS PETG (101-103: 5 PERIMETERS, 100% INFILL, EPOXY-COATED) · OTHERS PLA/PETG · NO SUPPORTS
SOURCE: replicator/cad/water_worm/water_worm_mini.scad · CHECKS: tools/w1_checks.py, w1_leak_check.py

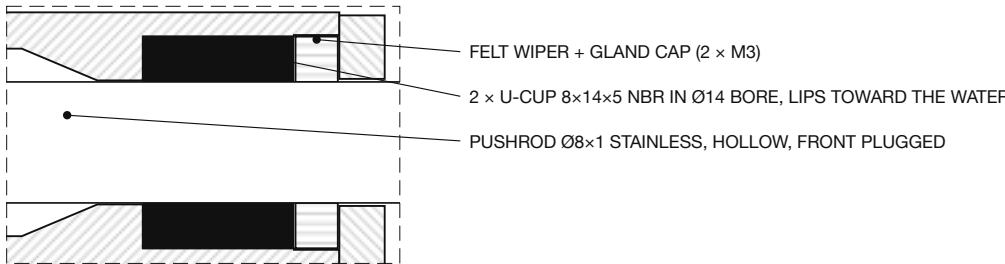
TITLE WATER WORM W1 · MINI TEST MODEL				REV 2.3	SHEET 1 / 4
DRAWING GENERAL ARRANGEMENT				DWG NO. W1-GA-001	
SCALE 1:2	UNITS mm	DATE 2026-09-30	DESIGNED D. LEITENBERG		
STATUS PROTOTYPE DESIGN · NOT YET BUILT OR TESTED					PROJECTION 

SECTION A-A · ON THE WORM AXIS

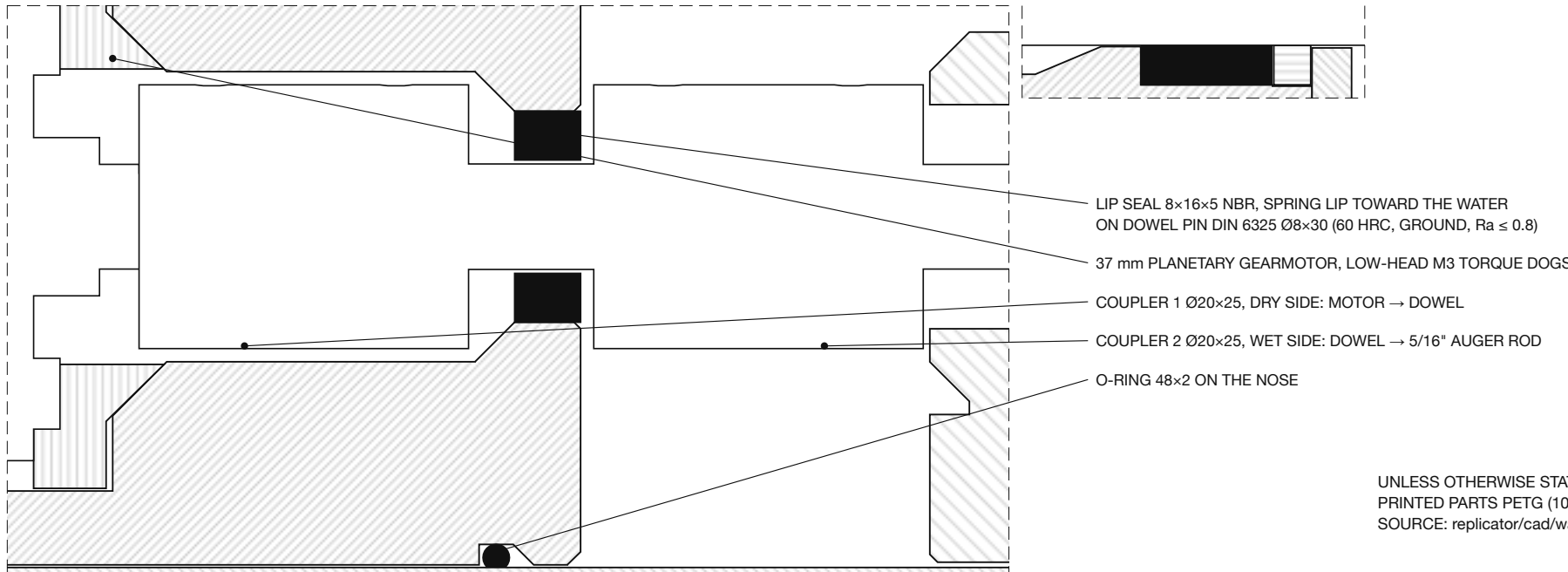
SCALE 1:2 · SHAFTS, SCREWS AND BOUGHT PARTS NOT SECTIONED



DETAIL C · PUSHROD GLAND · 2:1



DETAIL B · AUGER SEAL AT THE NOSE · 2:1

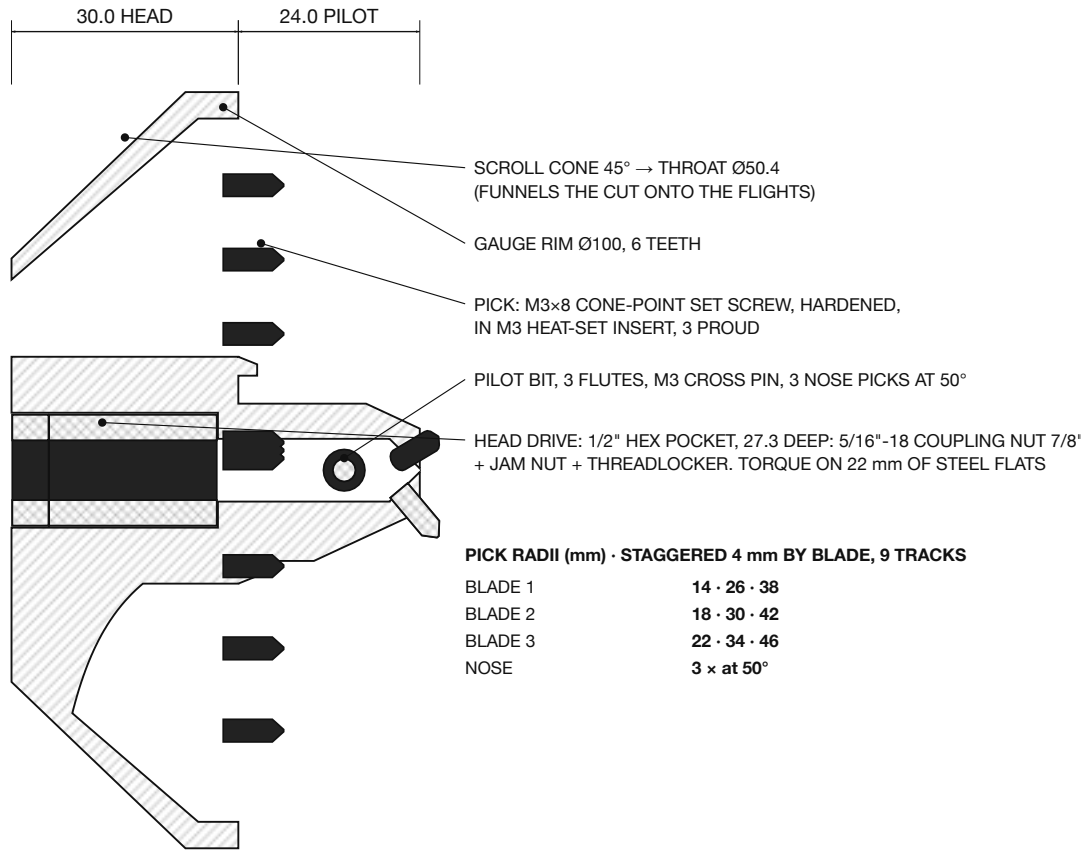
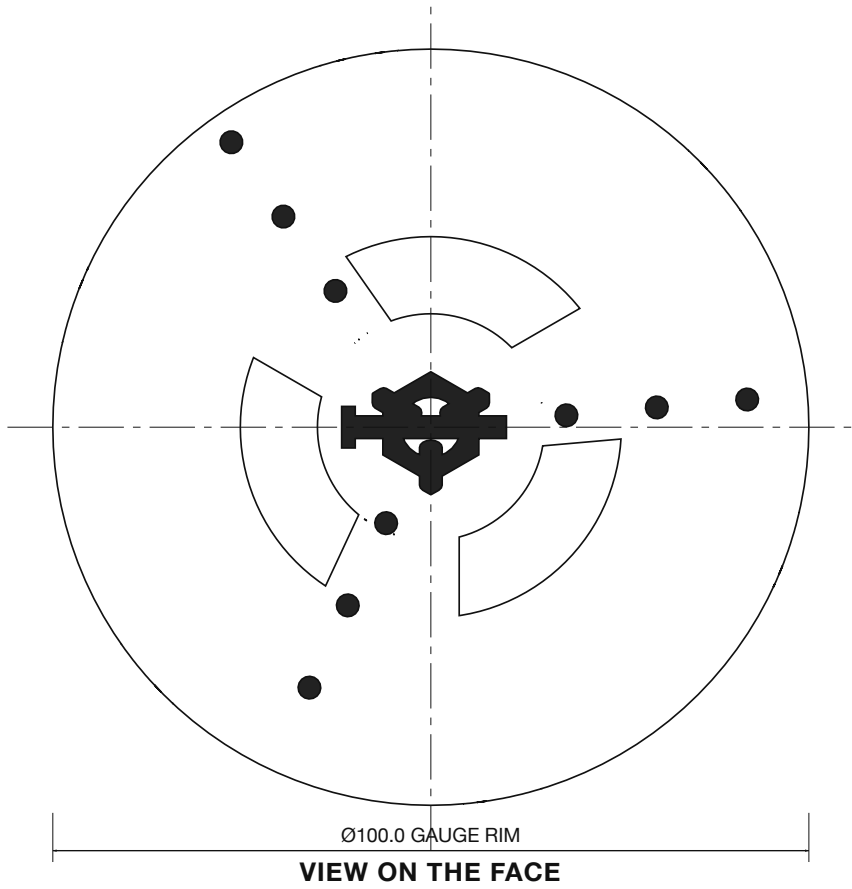


UNLESS OTHERWISE STATED: DIMENSIONS IN mm · PRINTED PARTS ±0.2 · DO NOT SCALE THE DRAWING
PRINTED PARTS PETG (101-103: 5 PERIMETERS, 100% INFILL, EPOXY-COATED) · OTHERS PLA/PETG · NO SUPPORTS
SOURCE: replicator/cad/water_worm/water_worm_mini.scad · CHECKS: tools/w1_checks.py, w1_leak_check.py

TITLE WATER WORM W1 · MINI TEST MODEL				REV 2.3	SHEET 2 / 4
DRAWING SECTION A-A · SEAL DETAILS				DWG NO. W1-SEC-002	
SCALE 1:2, 2:1	UNITS mm	DATE 2026-09-30	DESIGNED D. LEITENBERG		
STATUS PROTOTYPE DESIGN · NOT YET BUILT OR TESTED					PROJECTION 

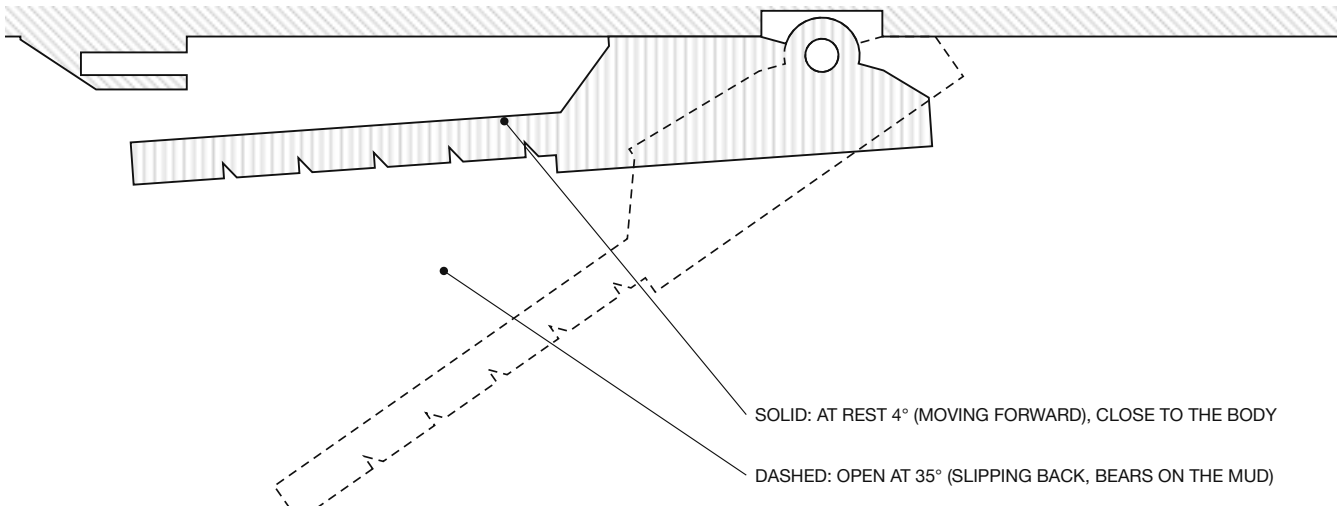
CROWN CUTTERHEAD (PART 117) · SCALE 1:1

AFTER THE ASTROWORM CROWN (astrowilly.scad aw_crown / aw_blade / aw_pick / aw_pilot_bit)



SECTION C-C

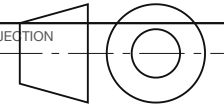
SETAE SHOE (PART 115) · SCALE 2:1 · SECTION ON THE BOTTOM ROW OF A (FRONT PIN, x 125)



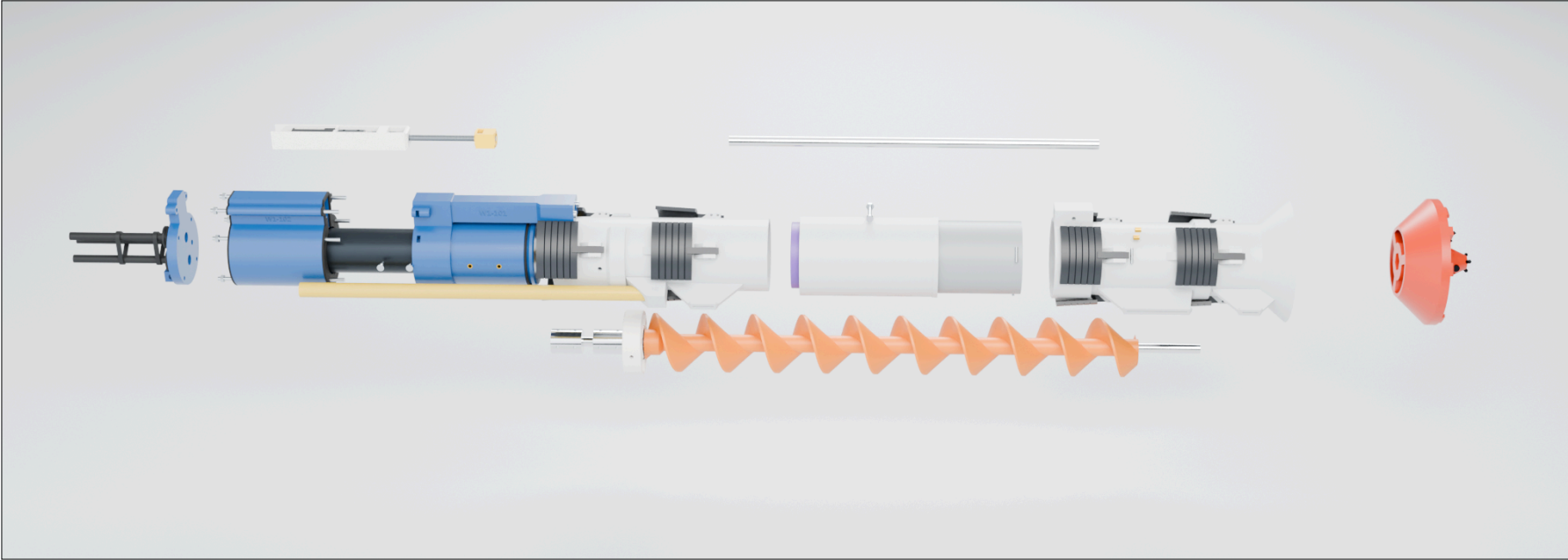
SHOE

Arc	80° round the worm axis
Length	32 (14-46 behind the pin)
Thickness	2.8
Inner radius at 0°	32.8
Rest / open	4° / 35°
Per section	6 (2 per row × 3 rows)
Pin	M2 × 12 hardened (M2's clevis)
Grooves	5, 1.0 deep, walls face the pivot
Prints	on its trailing edge, no supports

UNLESS OTHERWISE STATED: DIMENSIONS IN mm · PRINTED PARTS ±0.2 · DO NOT SCALE THE DRAWING
PRINTED PARTS PETG (101-103: 5 PERIMETERS, 100% INFILL, EPOXY-COATED) · OTHERS PLA/PETG · NO SUPPORTS
SOURCE: replicator/cad/water_worm/water_worm_mini.scad · CHECKS: tools/w1_checks.py, w1_leak_check.py

TITLE WATER WORM W1 · MINI TEST MODEL				REV 2.3	SHEET 3 / 4
DRAWING CROWN CUTTERHEAD · SETAE SHOE				DWG NO. W1-DET-003	
SCALE 1:1	UNITS mm	DATE 2026-09-30	DESIGNED D. LEITENBERG		
STATUS PROTOTYPE DESIGN · NOT YET BUILT OR TESTED					PROJECTION 

EXPLODED VIEW · ASSEMBLY ORDER



RENDERED FROM THE CAD (BLENDER CYCLES) · LEFT TO RIGHT = ASSEMBLY ORDER: LID, VESSEL, ACTUATOR + MOTOR, CASING A, AUGER, TELESCOPE, B, CROWN

PRINTED PARTS (MEASURED FROM THE STLs)

ITEM	NAME	QTY	MATERIAL	SIZE
101	Vessel, front (nose, hump, gland)	1	PETG, 100%, epoxy	77×58×124
102	Vessel, rear	1	PETG, 100%, epoxy	77×58×74
103	Lid (2 × WetLink M10, M5 air)	1	PETG, 100%, epoxy	77×58×20
104	Motor seat ring	1	PLA/PETG	41×38×10
105	Actuator sled	1	PLA/PETG	106×16×15
106	Motor spacer	1	PLA/PETG	37×37×15
107	Carriage (M5 nut)	1	PLA/PETG	12×12×16
108	Gland cap	1	PLA/PETG	17×24×3
110	Casing A, rear	1	PLA/PETG	66×69×58
111	Casing A, front (ports)	1	PLA/PETG	65×91×124
112	Collar (pushrod lug)	1	PLA/PETG	77×64×30
113	Sleeve + mouth ring	1	PETG	65×65×109
115	Setae shoe	12	PETG	12×46×53
116	Gasket template	1	PLA/PETG	58×77×1
117	Crown cutterhead	1	PETG	100×100×54
118	Fit coupon	1	PLA/PETG	84×52×6
2	Casing B, rear (M2)	1	PLA/PETG	68×91×108
19	Casing B, front (M2)	1	PLA/PETG	98×98×74
3	Pad, pipe tests (M2)	6	PLA/PETG	100×32×7
4	Link, pipe tests (M2)	16	PLA/PETG	19×5×5
6	Spigot (M2)	1	PLA/PETG	58×58×110
8	Flight (M2)	4	PLA/PETG	48×48×80
17	First flight (M2)	1	PLA/PETG	48×48×80
9	Rake head, optional (M2)	1	PLA/PETG	91×83×38
13	Thrust spider (M2)	1	PLA/PETG	52×52×15
24	Liner (M2)	1	PLA/PETG	53×53×84

BOUGHT PARTS

ITEM	NAME	QTY	MATERIAL	NOTE
B1	Gearmotor 37 mm planetary, 12 V ~15 rpm	1	bought	measure: can, shaft, PCD
B2	N20 encoder gearmotor 100-150 rpm	1	bought	
B3	Rigid coupler Ø20×25	2	aluminium	dry + wet
B4	Dowel pin DIN 6325 Ø8×30	1	steel 60 HRC	seal shaft
B5	Rotary shaft seal 8×16×5	1	NBR	
B6	U-cup rod seal 8×14×5	2	NBR	
B7	WetLink penetrator M10	2	Blue Robotics	size to jacket
B8	Push-in fitting M5, 4 mm tube	1		air line
B8a	Check valve, 4 mm tube	2		at the lid + at the pump
B9	O-ring 48×2	1	NBR	nose
B10	Pushrod tube Ø8×1×287	1	stainless	polished
B11	Lead screw M5×111 + nuts, F5-10M x2	1	steel	
B12	Auger rod 5/16"-18 x 481	1	ASTM A193 B7	not zinc rod
B13	Tie rod M3×100	6	steel	
B14	Heat-set insert M3	13	brass	
B15	Pick: M3×8 cone-point set screw	12	hardened	
B16	Gasket sheet 1.5 mm	2	neoprene	PART 116 template
B17	Hose 8×12	2	PVC/silicone	slurry, water

UNLESS OTHERWISE STATED: DIMENSIONS IN mm · PRINTED PARTS ± 0.2 · DO NOT SCALE THE DRAWING
PRINTED PARTS PETG (101-103: 5 PERIMETERS, 100% INFILL, EPOXY-COATED) · OTHERS PLA/PETG · NO SUPPORTS
SOURCE: replicator/cad/water_worm/water_worm_mini.scad · CHECKS: w1/w1_checks.py, w1_leak_check.py

TITLE				REV	SHEET
WATER WORM W1 · MINI TEST MODEL				2.3	4 / 4
DRAWING				DWG NO.	
EXPLODED VIEW · PARTS LIST				W1-BOM-004	
SCALE	UNITS	DATE	DESIGNED		
NTS	mm	2026-09-30	D. LEITENBERG		
STATUS				PROJECTION	
PROTOTYPE DESIGN · NOT YET BUILT OR TESTED				